PROJECT GUIDE



NATIVE LANGUAGE SYNC

Complete Architecture, Integrations, Operations & User Guide

Private multilingual translation workspace
Cloudflare Workers + Groq AI + GitHub

Prepared for: Project owner and technical reviewers

Documentation date: 11 November 2026 (Asia/Kolkata)

Source snapshot: GitHub `6bbbc77d8013` (main branch)

Source package label: 4.5.0; later layout/UI refinements are recorded in Git history. Worker/API reports 4.4.0.

Production URL: native-language-sync-free.native-translator.workers.dev

Scope / evidence statement

The architecture, features, filenames and limits described here were inspected in the connected GitHub source at the stated commit. User-confirmed deployment and Cloudflare/Groq connections are distinguished from settings that cannot be independently viewed. No passwords, API keys or private account identifiers are included.

00. Guide at a glance

This PDF is the project handover document. It is designed for daily use, future maintenance, audits and sharing with a developer without exposing private credentials.

SECTION	TOPIC
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Three things to remember

(1) Translation text goes from your browser to your Cloudflare Worker, then to Groq. (2) Passwords and API keys must stay in Cloudflare encrypted Secrets, never GitHub. (3) GitHub pushes trigger Cloudflare builds; verify deployment status after every change.

01. Executive project overview

Native Language Sync is a **password-protected, browser-based translator** focused on meaning-preserving multilingual and mixed-language input. The source is published in a GitHub repository; the running website is delivered by a Cloudflare Worker; Groq processes translation requests using a Qwen multilingual model.

ATTRIBUTE	IMPLEMENTATION
Application	Native Language Sync / private translation workspace
Primary goal	Any-language input → corrected English → selected language output
Frontend	React 19 + TypeScript + Vite 6 + CSS + Lucide icons
Backend	Cloudflare Worker with authenticated /api/translate endpoint
AI provider	GroqCloud API; Qwen 3.8 27B (model ID in source)
Hosting / URL	Cloudflare Workers and workers.dev subdomain
Source control	GitHub: KAJITH95/native-language-sync, main branch
Persistent app database	None evidenced in source
Saved history	Browser localStorage, up to 25 recent entries
Business model	Uses provider free tiers, subject to limits and policy changes

What it is / what it is not

- **Is:** private three-panel writing and translation tool; text-first with dynamic destination map.
- **Is:** AI-assisted English correction, tone selection and complete email drafting.
- **Is not:** an email sending service; Email mode prepares text for copy/paste only.
- **Is not:** a ChatGPT Plus API integration, Vercel AI Gateway or Gemini production backend.
- **Is not:** guaranteed unlimited, error-free or offline AI translation.

02. System architecture and request flow

Conceptual architecture

WEB BROWSER	CLOUDFLARE WORKER	GROQ API
React / TypeScript UI	Password + session checks	Qwen 3.8 27B
3 input/output panels	Authenticated /api/translate	English + localized JSON

Request direction: Browser → Worker (checks password session) → Groq model → Worker JSON → Browser fills English and destination cards.

Step-by-step request lifecycle

- User opens the workers.dev URL. The Worker checks that SITE_PASSWORD and GROQ_API_KEY are configured.
- If not signed in, the Worker presents a local password form. A successful sign-in sets a secure signed session cookie.
- User enters any-language or code-mixed text; the UI waits approximately 1,250 ms after typing pauses.
- Browser POSTs text, selected target code and writing tone to /api/translate on the same website.
- The Worker enforces authentication, same-origin protections, body limits and basic per-instance request throttling.
- The server sends the content with system instructions to Groq (not from the browser directly).
- Groq returns JSON. The server validates the response and formats email fields when Email mode is active.
- Browser displays corrected English in Box 2 and translated output in Box 3; optional history is saved locally.

03. Frontend experience and UI

AREA	BEHAVIOUR
Box 1 – Original	Paste, type or dictate text. No source language selector. Input limited to 1,800 characters.
Box 2 – English	Shows AI-refined English. Modes: Standard, Formal, Casual and Email.
Box 3 – Localized	Country and language selectors plus floral atlas map and geographically positioned country pin.
Header	Brand globe/flower mark, appearance toggle, history and log out.
Utility actions	Copy text, browser voice playback, input microphone, paste, clear, download results.
Auto translation	Automatically starts after a 1.25-second typing pause; tone and target changes re-trigger.
History	Recent successful translations stored on the current device, not an account database.
Appearance	Premium floral background, botanical decorative SVG, light/dark theme; desktop viewport-fit layout.

One-screen layout design

The desktop CSS rule for **width at least 851 px and height at least 680 px** constrains the workspace to the viewport and moves long text scrolling inside input/output areas. Narrow or short screens use normal page scroll to avoid inaccessible controls. Browser zoom, toolbars and OS text scaling affect the actual fit.

World atlas / floral graphics

The destination map is a bundled SVG-style vector graphic with locally defined land path and preconfigured coordinate mappings for **40 selectable countries**. The selected country drives the pin; the selected language appears in the label. It is not a geolocation service, live GPS or paid mapping API.

Accessibility and browser considerations

- Keyboard actions and focus outlines are present for key interactive controls.
- Voice input depends on SpeechRecognition/webkitSpeechRecognition, browser support and microphone permission; speech output uses speechSynthesis.
- Dark mode is stored locally; reduced-motion preferences disable selected atlas animations.
- Text overflow is intentionally scrollable within the editor/output; this is not unlimited single-page height.

04. AI meaning-first translation and Email mode

Two-stage translation

Stage 1: detect/interpret the intent of any language, mixed script or transliteration and produce grammatically correct English. **Stage 2:** translate that refined English into the selected destination language. The model is instructed not to invent information, names or events; this is a prompt objective, not a mathematical guarantee.

STYLE	EXPECTED OUTPUT
Standard	Clear corrected English with original formality preserved.
Formal	Polished, professional and respectful English without added facts.
Casual	Friendly conversational English with corrected grammar.
Email	Full draft with subject, salutation, body, closing and signature placeholder; translated email retains the same overall structure.

Email mode example

```
Input (Tanglish): Please candidate feedback email ready pannunga.
Email English output (illustration):
Subject: Candidate Evaluation Feedback

Hello,

I would like to share my feedback on the candidate.
The candidate has 10 years of experience.
Murex knowledge requires improvement.

Best regards,
[Your Name]
```

The server requests structured email sections from the model and assembles a consistent, ready-to-copy format. Where a person or address is not provided, placeholders or neutral salutations are preferred. Email mode does **not** send any email.

Service-specific implementation

- Groq endpoint: `https://api.groq.com/openai/v1/chat/completions`.
- Default model: `qwen/qwen3.8-27b`; JSON object mode, non-reasoning setting, max output tokens 2,600, temperature 0.15.
- Translation request timeout: approximately 27 seconds; input maximum 1,800 characters.
- Success can be cached in Worker memory for 10 minutes (up to 150 entries per warm instance), using text + language + tone as key.
- Model and provider may change; check Groq Console for availability before updating code.

05. External links, accounts and integrations

This is the complete list of external systems **identified in the current source or confirmed by the owner during setup**. “Used at runtime” is different from “used to develop or deploy.”

SERVICE / DOMAIN	ROLE	STATUS
GitHub `github.com`	Stores the public code repository, version history and PDF guide. Git push triggers Cloudflare build.	Connected (owner-confirmed)
Cloudflare `workers.dev`	Hosts website HTML/assets and server-side Worker; controls auth, API access and secrets.	Active (owner-confirmed)
GroqCloud `api.groq.com`	Runs Qwen AI for translation and email drafting. Request text is sent to Groq via Worker.	Active (translation observed by owner)

Google Fonts
`fonts.googleapis.com`
`fonts.gstatic.com`
	Loads DM Sans, Manrope and Noto Sans Tamil styles/fonts in the browser.	Referenced in index.html

Browser speech APIs	Microphone recognition and speech synthesis; capabilities depend on browser/platform.	Browser-dependent
Local browser storage	Stores theme, target country/language and recent translation history.	Used by frontend
Cloudflare browser URL	Provides public address but content/API require application password once correctly configured.	Public URL, gated access

Not connected to the current runtime

- **OpenAI/ChatGPT API:** not called by this app; the ChatGPT Plus plan is not the app’s AI backend.
- **Gemini / Google Generative AI:** older AI Studio version was replaced; not part of current production model path.
- **Vercel AI Gateway:** an earlier attempt; not used in the current Cloudflare deployment.
- **MyMemory translator:** belonged to an earlier implementation; not in this current Worker request path.
- **Email provider, database, analytics, payments or separate map API:** no corresponding integration found in the inspected source.

Data flow disclosure

Whenever you submit translation text, the backend sends that content to Groq. Cloudflare also processes HTTP requests. Browser speech recognition may use platform services, depending on the browser. Treat all externally processed text according to the relevant provider privacy policies.

06. Repository structure and technology stack

GitHub repository: `KAJITH95/native-language-sync`, branch `main`. The modern app is mirrored at the repository root and inside `cloudflare/` to support either configured Cloudflare build root. This duplication should be maintained carefully or consolidated later.

```
native-language-sync/
  src/
    App.tsx                   Main three-box React UI
    main.tsx                  React entry point
    styles.css                Premium theme + responsive one-screen CSS
    components/FloralAtlas.tsx  Dynamic atlas + flower/globe mark
    data/countries.ts         Destination list and languages
    data/worldAtlas.ts        Local world outlines + country pins
    types.ts                  TypeScript interfaces
  public/floral-world.svg     Decorative background (bundled asset)
  api/translate.js            Groq request, prompt, validation, email format
  worker/index.js             Auth/session/routing/asset protection
  tests/*.test.mjs            Mock-based Worker + atlas tests
  index.html                  App document and Google Fonts links
  vite.config.ts              Vite app and local translation middleware
  wrangler.jsonc              Cloudflare Worker/static asset config
  package.json                Dependencies and npm commands
  cloudflare/                 Mirrored deployable app source
  docs/                       Project documentation (this PDF)
```

Main frameworks, libraries and scripts

COMPONENT	SOURCE / ROLE
React 19 + React DOM 19	UI rendering, state/hooks and virtual DOM
TypeScript 5.8	Typed React source and build-time checking
Vite 6 + React plugin	Frontend build; outputs dist/ assets
Lucide React 0.546	Icons in user interface
Wrangler 4	Cloudflare Worker deployment/dev tooling
Node.js / npm	Dependency installation and local test/build scripts
Web APIs	localStorage, clipboard, Blob/download, speech recognition/synthesis, fetch, crypto

Note on versions: package.json currently says 4.5.0 while later CSS-only and UI cleanup commits were added; Worker/API constants report 4.4.0. Use the Git commit SHA as the precise source version. These displayed labels are not guaranteed to be synchronized.

07. Production configuration, deployment and secrets

A. Existing deployment settings

SETTING	VALUE / CHECK
Cloudflare Worker name	native-language-sync-free
Production source	GitHub → KAJITH95/native-language-sync → main
Root directory	Either repo root or cloudflare/ (both currently mirrored); verify actual choice in Cloudflare Settings → Builds
Build command	npm run build
Deploy command	npx wrangler deploy
Worker entry	./worker/index.js
Static assets	./dist, binding ASSETS
Auth ordering	assets.run_worker_first = true
Compatibility date	2026-11-11
Public address	https://native-language-sync-free.native-translator.workers.dev/

B. Required runtime secrets — production

EXACT NAME	TYPE	PURPOSE
SITE_PASSWORD	Secret	Private website password. Current code requires 10+ characters.
GROQ_API_KEY	Secret	Groq authentication token; used only server-side.

Cloudflare: Workers & Pages → native-language-sync-free → Settings → Variables and Secrets → Production → Add variable → Secret → Deploy. Values are private; no copy should be published in GitHub or embedded in the PDF. Do not confuse **build-time** secrets with **runtime** secrets.

C. Automatic deployment flow

- Edit GitHub source and commit to main.
- Cloudflare Workers Builds detects the commit and runs npm run build.
- Wrangler deploy publishes the Worker and compiled frontend assets.
- Check Cloudflare Deployments / Builds and the active production version.
- Open the existing workers.dev URL in an Incognito window and test login + a real translation.

Important deployment caution

A green GitHub commit means the source was saved, not necessarily that Cloudflare promoted it. A “Success/Deployed” build means deployment succeeded, but app behaviour still needs browser verification.

08. Authentication, privacy, storage and security

Sign-in design

• POST /auth/login checks the submitted password against Cloudflare SITE_PASSWORD and issues a signed session cookie.

• Session cookie: `__Host-nls_session` with Secure, HttpOnly, SameSite=Strict, Path=/; validity approximately 24 hours.

• Worker cryptographically signs session expiry using a key derived from both secret values; changing either secret makes old sessions fail.

• POST /auth/logout clears the browser cookie. Note: no central server-side session revocation table is implemented.

• Worker checks signed-in status before static asset access and API requests; authentication runs before asset serving.

• POST login and translation requests include same-origin / Fetch Metadata checks. Excessive login/translation requests are throttled by in-memory IP counters.

Personal data / storage map

DATA	WHERE / HOW LONG
Text currently in editor	Browser React state while page is open; sent to Groq on translation.
Recent translations	Browser localStorage, up to 25 items; persists until browser data/history is cleared.
Theme and selected language	Browser localStorage, persists on that device/profile.
AI request payload	Cloudflare Worker sends to Groq API. Provider retention governed by provider policy.
Model response cache	Worker process memory (not durable database); 10-min TTL; per-instance.
Password and Groq key	Encrypted Cloudflare runtime secrets. Value not intentionally sent to frontend.
Potential platform logs	Cloudflare/Groq logging depends on their services and account configuration; verify separately.

Security status is not a formal audit

Password gating is useful for a personal project, but the source has not had a third-party penetration test. Per-instance rate limits do not guarantee global anti-abuse protection. Avoid translating highly sensitive credentials or regulated personal data without a separate security/privacy assessment.

09. Daily user guide (owner and family)

Start using the live site

- Open the saved workers.dev link in Chrome, Edge or another current browser.
- Enter the SITE_PASSWORD you set privately in Cloudflare (never share it in chat).
- In Box 1, paste/enter Tamil, Tanglish, English, Hindi, mixed language or another text language.
- Wait about 1.25 seconds after stopping typing; translation starts automatically.
- Read the corrected English in Box 2; choose Standard, Formal, Casual or Email.
- Select a destination country and language in Box 3. Confirm the matching atlas country pin.
- Copy, play speech or download the result. For long text, scroll inside the text box.
- Click Log out when you finish; use History if you need a recent translation on the same device.

Useful examples

```
INPUT:  naan multiple times try panniten, still work aagala
ENGLISH: I have tried multiple times, but it still does not work.

INPUT:  candidate feedback-a email format la ready pannunga
EMAIL:  Choose Email → Subject / greeting / body / sign-off

DESTINATION: Select South Korea → Korean
ATLAS:  Country pin moves to South Korea and label updates.
```

Privacy and quality checklist

- Review proper nouns, dates, numbers, technical terms and negations before sending a translated message.
- Email mode drafts text; copy it into your actual mail client, add recipients and send manually.
- Voice/mic features may require microphone permission and may not work in every browser.
- If a translation seems inaccurate, edit the original or switch writing style and review again.
- Bookmark the URL; there is no desktop installation or ZIP required for daily use.

10. Local development and testing guide

Prerequisites

- Git access to the repository, an up-to-date Node.js LTS/npm installation, and optional Wrangler login for deploying.
- For real local Groq translation, configure only a local ignored .dev.vars or .env file, not GitHub secrets.

Typical workflow (from chosen build directory)

```
git clone https://github.com/KAJITH95/native-language-sync.git
cd native-language-sync
npm install
npm test
npm run build
npm run dev
```

Vite local preview generally starts on port 5173. For local Worker behaviour, use `npm run dev:worker` after dependencies/build are ready. Avoid committing generated `dist/`, `node_modules/` or local secret files.

Local secrets example (do not commit)

```
# .dev.vars (example names only; DO NOT commit real values)
SITE_PASSWORD=replace_with_your_own_very_strong_password
GROQ_API_KEY=replace_with_the_actual_private_key
```

The test suite contains **19 core Worker/AI/email tests plus 4 floral atlas tests** as source test cases. They mock the AI provider and verify key routes, auth behaviours, formats, map data and CSS contract. The documentation audit did not rerun live model requests; source tests do not guarantee future provider availability.

11. Free tiers, usage and monitoring

LIMIT AREA	CURRENT INFORMATION / IMPLICATION
Groq Free quotas	Groq publishes model-specific RPM, RPD, TPM and TPD; Qwen 3.8 27B documentation listed 30 RPM, 1,000 RPD, 8,000 TPM and 200,000 TPD when reviewed. Account dashboard is authoritative.
Groq throttling	HTTP 429 if any provider rate threshold is exceeded; model availability and limits can change.
Cloudflare Workers Free	Cloudflare documentation listed 100,000 requests/day and other CPU/memory restrictions; limits are account/plan dependent.
App-level translation rate	Worker code limits ~12 translations/minute per IP per active instance. Login rate limit ~8 attempts/10 minutes per IP per instance.
Input size / provider tokens	UI text ≤ 1,800 characters; model max response tokens 2,600; longer email may be truncated if response exceeds budget.
Automatic translation	Debounce ~1.25 seconds; many pauses/changes can still consume many Groq API calls.
Paid upgrade	Never upgrade from a Free account without reviewing terms, usage and optional billing settings.

Where to see usage

Groq Console — Usage (tokens/requests)

Groq Console — account rate limits

Cloudflare Dashboard — Worker Metrics / Deployments / Settings

For one-person, light daily use, the architecture is practical within free quotas, but **unlimited daily use is not promised**. The model pricing page may display paid rates even if you are operating inside a Free plan; use your console to confirm account tier and billing status.

12. Troubleshooting playbook

SYMPTOM	LIKELY CAUSE / ACTION
Security setup required	Production SITE_PASSWORD or GROQ_API_KEY missing; password must be at least 10 characters. Add both as runtime Secrets and Deploy.
Incorrect password	Re-enter your private password; if needed reset SITE_PASSWORD in Cloudflare (existing sessions become invalid).
Cross-site requests not permitted	Same-origin security check blocked request; use current HTTPS site normally (not an embedded page); confirm latest Worker deployed.
Login works, translation fails	Check Groq API key, account/model availability, usage limits, DevTools network and Worker logs.
HTTP 401 / login redirect	Session absent/expired, browser cookies blocked or changed secret values; sign in again.
HTTP 429	Provider free-tier throttle or Worker per-instance limit; reduce request frequency and wait for the applicable reset.
HTTP 503 / model unavailable	Provider not reachable, invalid secret or selected model withdrawn; review logs and Groq model availability.
HTTP 502 / invalid JSON	Model output did not match required JSON structure. Retry shorter text and inspect Worker logs.
Old UI after Git push	Cloudflare build root wrong, build not promoted, browser cache or old deployment. Check active commit and hard refresh.
No automatic deployment	Check Cloudflare Settings → Builds repository, main branch, build/root/deploy commands.
Mobile scrolling	Expected by responsive design; desktop only uses no-page-scroll above size threshold.
Voice button not working	Browser speech API/permission/language support; use Chrome/Edge and check microphone settings.

13. Ongoing maintenance, backup and rollback

Safe update procedure

- Back up or tag a known-good GitHub commit before structural changes.
- Apply changes to the correct source files; if mirroring remains, update both root and cloudflare/ copies.
- Run npm test and npm run build; review that neither source nor build output contains credentials.
- Push to main and monitor the resulting Cloudflare build, deployment and active version.

- Test password/login, mixed-language translation, Email mode, atlas country pin and logout in the browser.
- If a deployment fails, review build logs; revert the breaking commit and redeploy instead of sending ZIP files.

Backup responsibilities

- **Code:** GitHub history holds the source; optional clone to another secure drive.
- **Secrets:** keep keys/passwords in a private password manager. They are not recoverable from published source or this PDF.
- **History:** translation history is stored locally per browser. Clearing browser data can remove it; it is not backed up to GitHub.
- **Provider accounts:** preserve Cloudflare and Groq account access/recovery methods and review permissioned GitHub integrations.

Suggested monthly review

- Check Groq usage and current Free limits; check the model ID is still offered.
- Check Cloudflare Workers limits, build errors and deploy history.
- Update dependencies carefully and run tests; review security issues.
- Rotate exposed credentials immediately and redeploy; review GitHub for accidental secret commits.

14. Known limitations and future improvements

AREA	CURRENT LIMITATION / RECOMMENDATION
Translation accuracy	AI may alter a nuance or terminology; review critical text. Add controlled regression examples for your use cases.
Single-person security	Shared password and in-memory rate limits are not enterprise IAM. Consider Cloudflare Access or passkeys for stronger access policies.
Privacy	Groq and platform services receive request content/metadata; inspect privacy terms before using confidential documents.
Availability	Free API quotas and preview model can change; define a fallback model/provider only after testing security and cost.
History	Device-local storage only. Consider opt-in encrypted export/backup instead of a mandatory cloud database.
Build structure	Two mirrored source trees are maintenance risk. Consolidate to one root when Cloudflare build settings are fully standardized.
Documentation	Code/package/Worker version numbers not fully synchronized. Adopt release tags and maintain one changelog.
Email output	Email draft is generated, not delivered. Recipient/account integration would be an independent future feature.
Maps	Country pin is based on the selected country, not the language’s geographic origin or precise city location.

Recommended next engineering steps

(1) Keep one canonical source folder. (2) Add continuous GitHub CI test/build checks. (3) Use stronger global abuse protection if the public URL is broadly shared. (4) Add automated end-to-end browser tests. (5) Track release tags and verified source/version labels.

15. Reference links and final review checklist

Official product and documentation links below are included as clickable PDF links. Limits and product terms are subject to change.

Production website — https://native-language-sync-free.native-translator.workers.dev/

Source repository — https://github.com/KAJITH95/native-language-sync

Source snapshot commit — https://github.com/KAJITH95/native-language-sync/commit/6bbbc77d8013abadcc24d630a99c97407da26143

Cloudflare Dashboard — https://dash.cloudflare.com/

Cloudflare Git integration — https://developers.cloudflare.com/workers/ci-cd/builds/git-integration/

Cloudflare build setup — https://developers.cloudflare.com/workers/ci-cd/builds/configuration/

Cloudflare encrypted Secrets — https://developers.cloudflare.com/workers/configuration/secrets/

Cloudflare Workers Free limits — https://developers.cloudflare.com/workers/platform/limits/

Groq Console — API Keys — https://console.groq.com/keys

Groq Console — Usage — https://console.groq.com/usage

Groq Console — Limits — https://console.groq.com/settings/limits

Groq published rate limits — https://console.groq.com/docs/rate-limits

Groq Qwen 3.8 model page — https://console.groq.com/docs/model/qwen/qwen3.8-27b

Google Fonts — https://fonts.google.com/

Project owner acceptance checklist

- ☐ I can open the website and sign in without showing my password to anyone.
- ☐ Original text, English rewrite, selected destination translation and map pin all work.
- ☐ Email mode returns Subject / greeting / body / closing / signature text.
- ☐ GitHub push triggers Cloudflare Builds and deployment status is visible.
- ☐ Both SITE_PASSWORD and GROQ_API_KEY exist only as Cloudflare Secrets.
- ☐ I know how to review Groq usage and where to diagnose deployment errors.
- ☐ I have a copy of this PDF and a known-good Git commit for rollback.

Documentation prepared from the inspected GitHub source on 11 November 2026. This is an architecture and owner-operation handover, not a claim of formal security certification or a live production performance audit.